

## ***Smoking and lung cancer risk in American and Japanese men: an international case-control study.***

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Rates of lung cancer in American men have greatly exceeded those in Japanese men for several decades despite the higher smoking prevalence in Japanese men. It is not known whether the relative risk of lung cancer associated with cigarette smoking is lower in Japanese men than American men and whether these risks vary by the amount and duration of smoking. To estimate smoking-specific relative risks for lung cancer in men, a multicentric case-control study was carried out in New York City, Washington, DC, and Nagoya, Japan from 1992 to 1998. A total of 371 cases and 373 age-matched controls were interviewed in United States hospitals and 410 cases and 252 hospital controls in Japanese hospitals; 411 Japanese age-matched healthy controls were also randomly selected from electoral rolls. The odds ratio (OR) for lung cancer in current United States smokers relative to nonsmokers was 40.4 [95% confidence interval (CI) = 21.8-79.6], which was >10 times higher than the OR of 3.5 for current smokers in Japanese relative to hospital controls (95% CI = 1.6-7.5) and six times higher than in Japanese relative to community controls (OR = 6.3; 95% CI = 3.7-10.9). There were no substantial differences in the mean number of years of smoking or average daily number of cigarettes smoked between United States and Japanese cases or between United States and Japanese controls, but American cases began smoking on average 2.5 years earlier than Japanese cases. The risk of lung cancer associated with cigarette smoking was substantially higher in United States than in Japanese males, consistent with population-based statistics on smoking prevalence and lung cancer incidence. Possible explanations for this difference in risk include a more toxic cigarette formulation of American manufactured cigarettes as evidenced by higher concentrations of tobacco-specific nitrosamines in both tobacco and mainstream smoke, the much wider use of activated charcoal in the filters of Japanese than in American cigarettes, as well as documented differences in genetic susceptibility and lifestyle factors other than smoking.